Margherita Firenze

MIT PhD Student

Massachusetts Institute of Technology
32 Vassar Street
Cambridge, MA 02139

mfirenze@mit.edu
https://github.com/mafirenze
mafirenze

Research Overview

I am a third-year PhD student in the Electrical Engineering and Computer Science Department at MIT. My research focuses on multi-scale, rapid registration methods with applications in fetal MRI 2D to 3D reconstruction. I am particularly interested in building deep learning tools that exploit multi-modal data, physics-based constraints, and expand what is possible to extract from conventional imaging and sensor data.

Research Areas: *computer vision*, *deep-learning*, *super-resolution*, *MRI*, *image analysis*,

Education

Massachusetts Institute of Technology *2023 - Present*
Ph.D. in Electrical Engineering & Computer Science, GPA 5.00/5.00 Advisor: Polina Golland

Columbia University *2019-2023*
B.S. in Electrical Engineering, *Magna Cum Laude* Advisor: Christine Hendon

Research Experience

Computer Science and Artificial Intelligence Laboratory, MIT *2023 - Present*

- PhD student in the Medical Vision Group building 2D to 3D reconstruction models for fetal brain MRI reconstruction, using a deep-learning synthetic data approach
- Coursework in machine learning, computer vision, symmetry for machine learning, and computational imaging, human reproductive biology
- Collaborating with inter-disciplinary hospital teams to advance fetal MRI analysis

Department of Electrical Engineering, Columbia University *2021 - 2023*

- Undergraduate researcher in the Structure Function Laboratory developing novel optical imaging techniques.
- Developed file automation scripts in Python and MATLAB to prepare dataset for deep-learning, processed over 800GB of image data
- Developed a Pytorch deep learning model to classify breast biopsy Optical Coherence Tomography (OCT) images as cancerous or non-cancerous

Bristol Myers Squibb (BMS), Translational Medicine – Imaging Intern *Jun 2022 – Aug 2022*

- Shadowed PET and CT imaging pre-clinical studies.
- Conducted an independent research project using mathematical modeling to predict the kinetics of central nervous system radiotracers in PET studies.

Fellowships and Awards

Highlight Poster, MIT-MGB AI Cures Conference 2025
Conference Travel Grant, CVPR conference 2024

Graduate Research Fellowship (*3 years of full funding*), NSF 2023
Optics and Photonics Education Scholarship, SPIE 2022

Publications

Under review

C1 **Margherita Firenze**, Sean I Young, Clinton Wang, Elfar Adalsteinsson, Hyuk Jin Yun, Patricia E. Grant, Kiho Im, Polina Golland "*Fast Convolutional Slice-to-Volume Reconstruction for Multi-Stack MRI.*" Computer Vision and Pattern Recognition (CVPR). Submitted.

C4 **Margherita Firenze**, Sean I Young, Clinton Wang, Elfar Adalsteinsson, Hyuk Jin Yun, Patricia E. Grant, Kiho Im, Polina Golland "*cSVR: Fast Convolutional Slice-to-Volume Reconstruction for Fetal Brain MRI.*" International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting & Exhibition. Submitted.

Abstracts

C2 R. Muthukrishnan, B. Billot, B. Gagoski, **M. Firenze**, M. Soldatelli, P. E. Grant, and P. Golland. "*3D fetal head pose estimation from MRI navigators with equivariant networks.*" International Society for Magnetic Resonance in Medicine (ISMRM) Annual Meeting & Exhibition. Accepted for poster presentation.

C3 **Margherita Firenze***, Yunhe Liu*, Diana Mojahed, Nisha Gandhi, Hanina Hibshoosh, Richard Ha, and Christine Hendon. "*Optical coherence tomography texture analysis and classification of breast biopsies.*" Advanced Biomedical and Clinical Diagnostic and Surgical Guidance Systems XXI, SPIE Photonics West. Accepted for oral presentation, January 2023.

C4 Nisha Gandhi, Diana Mojahed, **Margherita Firenze**, Hua Guo, Hanina Hibshoosh, Richard Ha, and Christine Hendon. "*Deep learning to identify breast disease in an 87-patient clinical study of breast core biopsies to provide rapid biopsy evaluation.*" Advanced Biomedical and Clinical Diagnostic and Surgical Guidance Systems XX, SPIE Photonics West. Accepted for oral presentation, January 2022. https://doi.org/10.1117/12.2610196

C5 **Margherita Firenze**, Diana Mojahed, and Christine P. Hendon. "*Image Processing Pipeline to Classify Patches of Interest in OCT Biopsy Dataset.*" Biomedical Engineering Society (BMES) Annual Conference, 2021. Oral presentation.

Teaching

AI Pioneers (Middle School) Summer 2021, 2023
Cohort Instructor, Inspirit AI

Fundamentals of Computer Systems (Undergraduate) Spring 2022, 2023
Course Assistant, Columbia University

Service

PhD Applications Reviewer, MIT EECS department 2025
Federal Affairs Committee Member, MIT Graduate Student Council 2025
Environmental Chair, Social Chair, Edgerton House, MIT 2024, 2025
Application Mentor, Graduate Application Assistance Program 2023, 2024
Outreach Chair, Graduate Women in EECS 2024
Orientation Leader, Columbia University 2022